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$$z^4 = 16$$

$$\Leftrightarrow z^4 = 16 \cdot \text{cis}(k \cdot 2\pi) \quad (k \in \mathbb{Z})$$

$$z_0 = 2 \cdot \text{cis}\left(\frac{0 \cdot 2\pi}{4}\right) = 2$$

$$z_1 = 2 \cdot \text{cis}\left(\frac{1 \cdot 2\pi}{4}\right) = 2i$$

$$z_2 = 2 \cdot \text{cis}\left(\frac{2 \cdot 2\pi}{4}\right) = -2$$

$$z_3 = 2 \cdot \text{cis}\left(\frac{3 \cdot 2\pi}{4}\right) = -2i$$

References